

```

1  /*****
2  * AUTHOR:      CYRIL ANDRE B. DURANGO
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
5  *
6  * -----
7  * PROJECT FILE TITLE: Problem_B.v
8  *
9  *****/
10 module Problem_B
11 (
12     input    [3:0] BCD_in,
13     output reg [8:0] BGraph_out
14 );
15
16     always @(*)
17     begin
18         casez (BCD_in)
19             4'd0 : BGraph_out = 9'b000000000 ;|
20             4'd1 : BGraph_out = 9'b000000001 ;
21             4'd2 : BGraph_out = 9'b000000011 ;
22             4'd3 : BGraph_out = 9'b000000111 ;
23             4'd4 : BGraph_out = 9'b000001111 ;
24             4'd5 : BGraph_out = 9'b000011111 ;
25             4'd6 : BGraph_out = 9'b000111111 ;
26             4'd7 : BGraph_out = 9'b001111111 ;
27             4'd8 : BGraph_out = 9'b011111111 ;
28             default: BGraph_out = 9'b111111111 ;
29         endcase
30     end
31
32
33
34 endmodule

```

Quartus Prime Lite Edition - Z:/HDL Program Files/Problem_B/Problem_B - Problem_B

File Edit View Project Assignments Processing Tools Window Help

Problem_B.v

Table of Contents

Flow Summary

Flow Status: Successful - Fri Dec 06 09:30:55 2024

Quartus Prime Version: 20.1.1 Build 720 11/11/2020 SJ Lite Edition

Revision Name: Problem_B

Top-level Entity Name: Problem_B

Family: MAX 10

Device: 10MS00AF484C7G

Timing Models: Final

Total logic elements: 9 / 49,760 (< 1 %)

Total registers: 0

Total pins: 13 / 360 (4 %)

Total virtual pins: 0

Total memory bits: 0 / 1,677,312 (0 %)

Embedded Multiplier 9-bit elements: 0 / 268 (0 %)

Total LUTs: 0 / 41 (0 %)

UPM blocks: 0 / 1 (0 %)

ADC blocks: 0 / 2 (0 %)

IP Catalog

Installed IP

Project Directory

No Selection Available

Library

Basic Functions

DSP

Interface Protocols

Memory Interfaces and Controllers

Processors and Peripherals

University Program

Search for Partner IP

Tasks

Compilation

Task

Compile Design

Analysis & Synthesis

Edit Settings

View Report

Analysis & Elaboration

Partition Merge

Netlist Viewers

Design Assistant (Post-Map)

I/O Assignment Analysis

View Report

Pin Planner

Filter (Place & Route)

Assembler (Generate program)

Timing Analysis

Messages

System (10) Processing (124)

10 Message

332102 Design is not fully constrained for setup requirements

332102 Design is not fully constrained for hold requirements

Quartus Prime Timing Analyzer was successful. 0 errors, 5 warnings

Running Quartus Prime EDA Netlist Writer

Command: quartus_eda --read_settings_files=off --write_settings_files=off Problem_B -c Problem_B

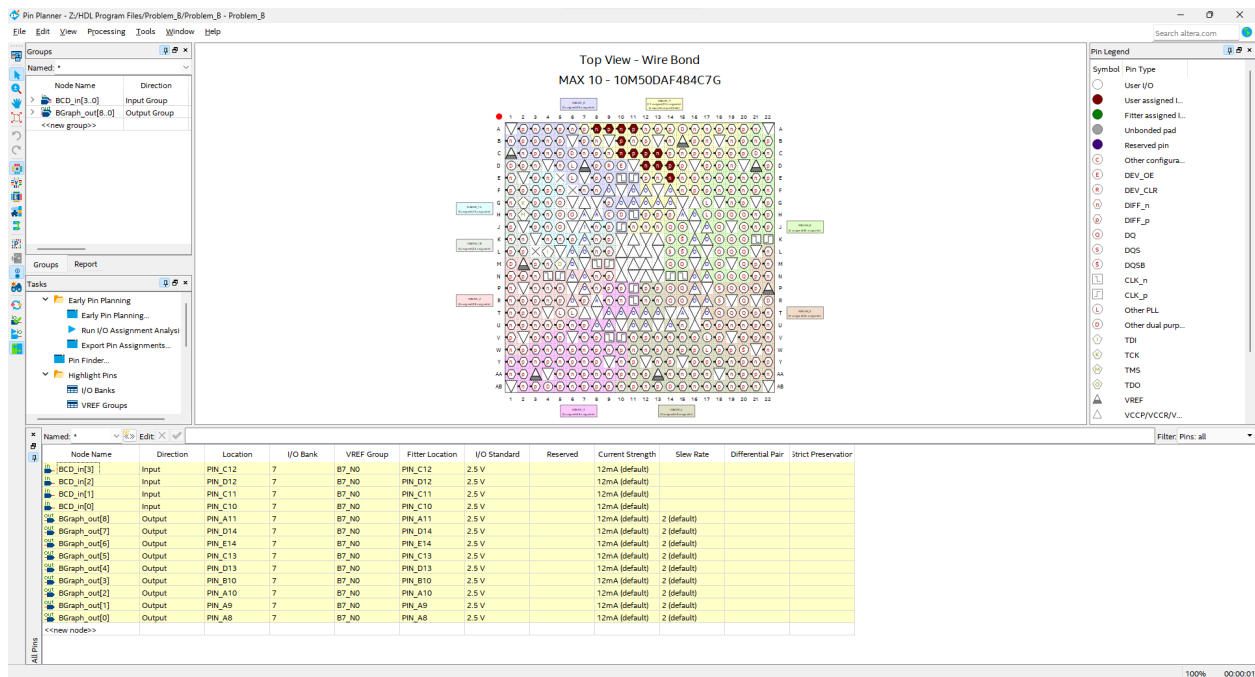
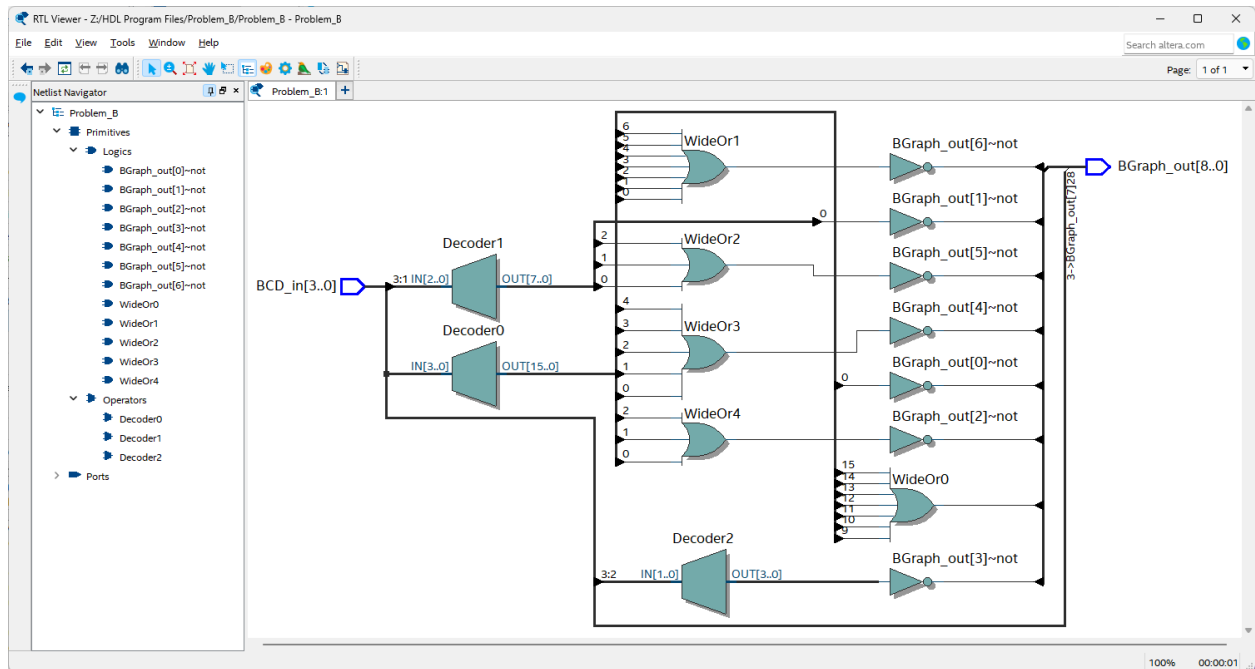
18236 number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your qsf to an appropriate value for best performance.

204019 generated File Problem_B.vo in folder "Z:/HDL Program Files/Problem_B/simulation/modelsim/" for EDA simulation tool

Quartus Prime EDA Netlist writer was successful. 0 errors, 1 warning

293000 Quartus Prime full compilation was successful. 0 errors, 13 warnings

100% 00:00:25



Name in Verilog File	FPGA Pin Assignment	Corresponding I/O Device
BCD_in[0]	PIN_C10	SW0
BCD_in[1]	PIN_C11	SW1
BCD_in[2]	PIN_D12	SW2
BCD_in[3]	PIN_C12	SW3
BGraph_out[0]	PIN_A8	LEDR0
BGraph_out[1]	PIN_A9	LEDR1
BGraph_out[2]	PIN_A10	LEDR2
BGraph_out[3]	PIN_B10	LEDR3
BGraph_out[4]	PIN_D13	LEDR4
BGraph_out[5]	PIN_C13	LEDR5
BGraph_out[6]	PIN_E14	LEDR6
BGraph_out[7]	PIN_D14	LEDR7
BGraph_out[8]	PIN_A11	LEDR8